

Radiomic Feature Extraction and Two-Year Outcome Classification in Non-Small-Cell Lung Cancer CT: A From-Scratch Descriptor Pipeline on the Lung1 Cohort

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Abstract—Radiomics assumes that quantitative descriptors computed from routine computed tomography carry prognostic information beyond what a radiologist records. This work tests that assumption end to end on the public NSCLC-Radiomics (Lung1) cohort using descriptor operators implemented from first principles rather than a pre-built radiomics library. A complete pipeline was built covering DICOM ingestion with geometric slice ordering, RTSTRUCT-based reconstruction of the GTV-1 gross tumour volume, boundary representation by Moore tracing, chain codes and Fourier descriptors, three-dimensional shape descriptors, and four families of texture descriptors comprising first-order statistics, the grey-level co-occurrence matrix, local binary patterns and Fourier spectral measures. Twenty-five candidate texture descriptors were subjected to a mask-perturbation stability test consisting of erosion and dilation by one and two voxels; nine descriptors survived a criterion of at most ten per cent change in at least eighty per cent of comparisons. The retained descriptors were used to predict a binary two-year survival endpoint in 419 patients under a single frozen stratified patient-level five-fold cross-validation partition, with feature selection and normalisation confined to the training folds. Six classifiers were compared: minimum distance, Gaussian Bayes, linear and radial-basis support vector machines, random forest, and a small convolutional network. Out-of-fold accuracy ranged from 0.568 to 0.597 and the area under the receiver operating characteristic curve from 0.528 to 0.595, against a majority-class rate of 0.598. The principal finding is therefore negative and is reported as such: after stability filtering, the retained descriptors carry only marginal prognostic signal for this endpoint in this cohort, and no classifier separated the classes meaningfully better than chance.

Index Terms—radiomics, computed tomography, non-small-cell lung cancer, texture analysis, grey-level co-occurrence matrix, local binary patterns, Fourier descriptors, feature stability, survival classification

I. INTRODUCTION

QUANTITATIVE image analysis of oncological computed tomography (CT), commonly termed radiomics, proposes that the spatial distribution of attenuation values inside a tumour encodes biological properties that correlate

with patient outcome [2]. The appeal is practical: the images already exist, they are acquired as part of standard care, and the analysis is non-invasive and inexpensive relative to molecular assays.

The claim, however, is only as strong as the descriptors that support it. A descriptor that changes substantially when the tumour contour is redrawn by a few voxels cannot carry stable biological meaning, and a classifier trained on such a descriptor will report an accuracy that does not generalise. Two methodological choices therefore dominate this study. First, every descriptor operator was implemented directly from its mathematical definition, following Chapters 11 and 12 of Gonzalez and Woods [1], rather than being called from an existing radiomics package; every reported number traces back to code written for this project. Second, the analysis protocol—censoring policy, endpoint definition, aggregation policy and cross-validation partition—was frozen before any classifier was trained, so that no decision could be retro-fitted to a favourable result.

The contributions of this paper are as follows.

- 1) A complete, reproducible pipeline from raw DICOM to out-of-fold predictions for the 422-patient Lung1 cohort, with from-scratch implementations of boundary, regional, three-dimensional shape and texture descriptors.
- 2) A mask-perturbation stability analysis of twenty-five texture descriptors, which eliminates sixteen of them, including several that are routinely reported in the radiomics literature.
- 3) A controlled comparison of six classifiers of increasing capacity on a single frozen cross-validation partition, with an explicit and quantitative negative result.

Section II describes the cohort. Section III details the pipeline. Section IV reports the results, and Section V discusses their interpretation and the limitations that bound them.

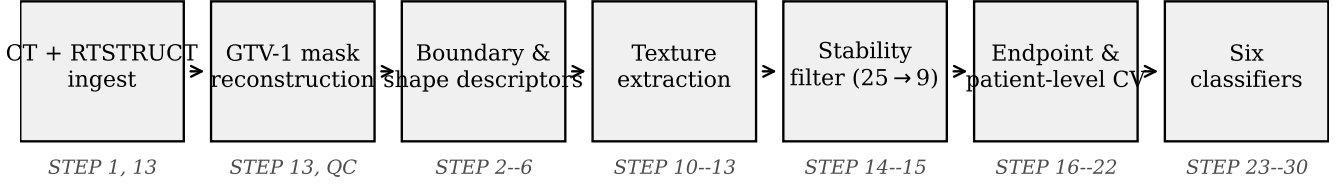


Fig. 1. Processing pipeline from raw DICOM to out-of-fold predictions. Italic labels give the numbered implementation stages of the accompanying code repository. Feature selection and normalisation occur strictly inside the training partition of each fold.

TABLE I
COHORT AND DATASET CHARACTERISTICS AS USED IN THIS STUDY

Property	Value
Source	TCIA, NSCLC-Radiomics (Lung1)
Patients in collection	422
Successful mask reconstruction	420 (99.53%)
Patients classified	419
Modality	Pre-treatment CT, DICOM series
Segmentation	GTV-1, from RTSTRUCT
In-plane spacing (LUNG1-001)	0.9766×0.9766 mm
Slice thickness (LUNG1-001)	3.0 mm
Endpoint	Two-year survival, 730-day threshold
Class 0 (death < 730 d)	251 patients (59.8%)
Class 1 (survival $\geq$ 730 d)	169 patients (40.2%)
Excluded at classification	LUNG1-024 (3 missing spectral features)

II. MATERIALS

A. Cohort

The NSCLC-Radiomics (Lung1) collection, distributed through The Cancer Imaging Archive [2], [3], provides pre-treatment CT studies for 422 patients with non-small-cell lung cancer, an expert delineation of the gross tumour volume distributed as DICOM RTSTRUCT objects, and an accompanying clinical table containing survival time, death-status indicator, age, sex, TNM category, overall stage and histology. Table I summarises the cohort as it was actually used in this study.

Two patients failed mask reconstruction and were excluded, leaving 420 patients with a complete descriptor vector. One further patient, LUNG1-024, was missing three spectral descriptors and was therefore excluded at the classification stage, so that all six classifiers were evaluated on an identical set of 419 patients. Keeping the evaluation set identical across classifiers is what makes the comparison in Section IV fair.

B. Endpoint

The prediction target is a binary two-year survival label defined at a threshold of 730 days: class 0 denotes death before 730 days and class 1 denotes survival to at least 730 days.

Class 0 accounts for 251 of the 420 patients, giving a majority-class rate of 59.8 per cent. This value is the reference against which every accuracy in Section IV must be read; a classifier that predicts class 0 for every patient already achieves it.

It should be recorded that an intermediate summary file produced during development reported a class balance of 251 versus 47 under a stricter censoring rule that excluded patients censored before 730 days. That rule was not adopted. The dataset used by every classifier reported here contains 251 class-0 and 169 class-1 patients, and the discrepancy is noted rather than silently discarded.

III. METHODS

Fig. 1 shows the pipeline. Each stage writes its own comma-separated result file and text report, so that any number in this paper can be traced to a specific stage output.

A. DICOM ingestion and mask reconstruction

Slices were never ordered by filename. For each series the slice normal was computed as the cross product of the two direction cosine vectors of `ImageOrientationPatient`, and slices were ordered by projecting `ImagePositionPatient` onto that normal. This ordering is correct for head-first and feet-first acquisitions alike and is robust to oblique geometry. Rescale slope and intercept were applied to obtain Hounsfield units before any intensity computation.

The GTV-1 contours were read from the RTSTRUCT object and rasterised into a binary three-dimensional mask on the CT voxel grid using the voxel-to-patient affine derived from the position, orientation and spacing attributes. Alignment between the reconstructed mask and the CT volume was verified visually for a sample of patients before the descriptor stages were run.

B. Boundary representation and shape descriptors

For each tumour-bearing slice the outer boundary was extracted by Moore boundary tracing, resampled to a fixed length of 64 points on a 64×64 grid, and encoded as an eight-connected Freeman chain code. The normalised first difference of the chain code, which is invariant to the choice of starting point and to rotation by multiples of 45° , was computed as

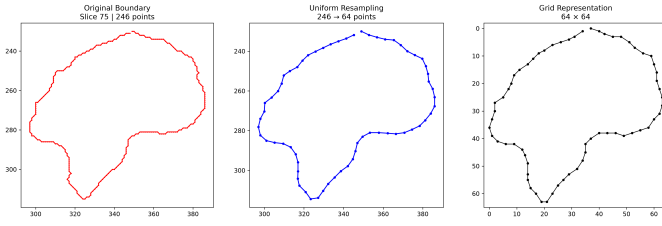


Fig. 2. Boundary processing for patient LUNG1-001, slice 75. From left to right: the outer boundary obtained by Moore tracing, the same boundary uniformly resampled to 64 points, and its representation on the 64×64 grid from which the eight-connected chain code is read.

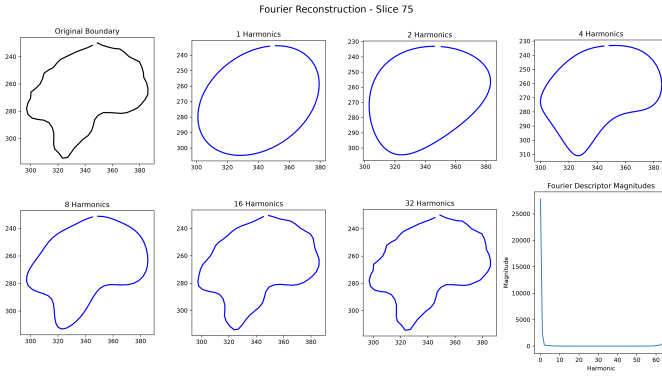


Fig. 3. Fourier-descriptor reconstruction of the same boundary at increasing harmonic truncation levels. Low-order harmonics carry the global shape; high-order harmonics restore local boundary irregularity.

the rotation-invariant descriptor [1]. Fig. 2 shows the extracted boundary, its resampled grid representation and the resulting code for a representative slice.

The boundary was also expressed as the complex sequence $z(n) = x(n) + jy(n)$ and transformed by the discrete Fourier transform to obtain Fourier descriptors. Reconstructions were computed from 1, 2, 4, 8, 16 and 32 harmonics to make explicit how much boundary detail each truncation level retains (Fig. 3). One harmonic recovers only the gross elliptical trend; eight to sixteen harmonics recover the major lobes and indentations; thirty-two harmonics recover most local irregularity. This behaviour is exactly what the descriptor is intended to expose, and it provides a compact, orderable measure of contour complexity.

A boundary straightness measure was computed per slice from the first-difference code. Across the 21 tumour-bearing slices of patient LUNG1-001 it varied between 0.327 and 0.675, being lowest in the mid-tumour slices where the contour is most lobulated and highest in the small apical and basal slices. No textbook definition of this quantity exists, so it is declared as an operational descriptor rather than a textbook one.

Regional descriptors—area, perimeter, compactness in the P^2/A form used by Gonzalez and Woods, circularity, and the seven Hu moment invariants—and three-dimensional descriptors—volume, surface area, sphericity and surface-to-volume ratio—were computed from the binary mask with

physical voxel spacing applied. These descriptors characterise the tumour geometry and are reported in the accompanying repository; they were not carried into the classification stage, which uses the texture descriptor set defined below.

C. Texture descriptors

Four families of texture descriptors were implemented and computed for every patient over the GTV-1 region:

- **First-order statistics** derived from the intensity histogram: mean, variance, smoothness, third moment, uniformity and entropy.
- **Grey-level co-occurrence matrix (GLCM)** descriptors: contrast, correlation, energy, homogeneity, entropy and maximum probability, computed from the normalised co-occurrence matrix as defined in [1], [4].
- **Local binary patterns (LBP)** [5]: the mean, variance, uniformity and entropy of the LBP code map.
- **Spectral descriptors** obtained from the two-dimensional Fourier spectrum: spectral energy, spectral entropy, dominant frequency, dominant orientation, fundamental period, and the mean and variance of the radial and angular spectral profiles.

This yields the twenty-five candidate descriptors listed in Table IV.

D. Stability filtering

A descriptor is only useful if it survives plausible variation in the tumour contour. Each patient mask was perturbed by morphological erosion and dilation of one and two voxels, giving four perturbed masks per patient, and every descriptor was recomputed on each. A descriptor was counted stable for a given patient and perturbation when its relative change was at most ten per cent, and it was retained overall when it was stable in at least eighty per cent of valid comparisons. With 420 patients and four perturbations this gives up to 1680 comparisons per descriptor.

The threshold pair (10 per cent, 80 per cent) was fixed before the results were inspected. It is a design decision rather than a derived optimum, and its consequences are discussed in Section V.

E. Endpoint, partition and leakage control

A single stratified patient-level five-fold partition was generated with random seed 42 and reused unchanged by all six classifiers. Table II gives the fold composition. Each patient contributes exactly one row and appears in exactly one validation fold, so no patient can leak between training and validation.

Feature selection and normalisation were performed inside the training partition of each fold only. Normalisation statistics were estimated on training patients and applied unchanged to validation patients. Where feature selection was used it was an ANOVA F -test ranking computed on the training partition alone.

TABLE II
STRATIFIED PATIENT-LEVEL FIVE-FOLD PARTITION

Fold	Training		Validation	
	Class 0	Class 1	Class 0	Class 1
1	200	135	51	34
2	201	135	50	34
3	201	135	50	34
4	201	135	50	34
5	201	136	50	33

TABLE III
CLASSIFIER CONFIGURATIONS

Classifier	Configuration	Features used
Minimum distance	Nearest class mean, Euclidean	9 (all retained)
Gaussian Bayes	Class-conditional normal, training priors	9 (all retained)
Linear SVM	Linear kernel	9, ranked per fold
RBF SVM	$C = 1.0$, $\gamma = \text{scale}$	5, ANOVA F per fold
Random forest	500 trees, Gini, balanced weights	5, ANOVA F per fold
Small CNN	20 epochs, batch 16, lr 10^{-3} ; conv-pool-conv-dense-dropout	9, as 3×3 map

F. Classifiers

Six classifiers of increasing capacity were compared (Table III). The first two were implemented from scratch following Chapter 12 of [1]: a minimum-distance classifier assigning each patient to the nearest class mean in the normalised descriptor space, and a Gaussian Bayes classifier estimating a class-conditional multivariate normal density with class priors taken from the training partition. The remaining four are standard implementations: a linear support vector machine, a radial-basis support vector machine, a random forest with balanced class weights, and a small convolutional network.

One point about the convolutional network must be stated plainly. Although tumour patches were cached for every patient, the network as evaluated does not operate on those patches. Its input is the nine retained descriptors arranged deterministically as a 3×3 matrix, passed through a convolution, max-pooling, second convolution, dense, dropout and output sequence. It is therefore a small network applied to a compact feature representation, not an image-based deep model, and it should not be read as evidence about what a patch-based network would achieve.

IV. RESULTS

A. Descriptor stability

Of the twenty-five candidate texture descriptors, nine met the retention criterion and sixteen were eliminated. Table IV gives the complete outcome and Fig. 4 shows it graphically.

Three observations follow from Table IV. First, entropy-like descriptors dominate the retained set: spectral, statistical, GLCM and LBP entropies are all retained, together with LBP

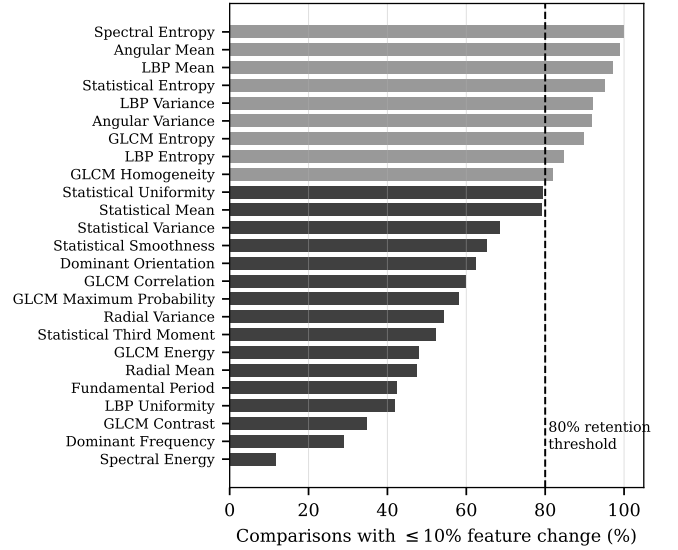


Fig. 4. Fraction of mask-perturbation comparisons in which each descriptor changed by at most ten per cent. Dark bars are retained descriptors, light bars eliminated ones; the dashed line is the eighty per cent retention threshold.

TABLE IV
MASK-PERTURBATION STABILITY OF ALL TWENTY-FIVE CANDIDATE TEXTURE DESCRIPTORS

Descriptor	Stable (%)	Mean Δ (%)	Max Δ (%)	Decision
Spectral entropy	99.88	2.40	13.6	Retained
Angular mean	98.93	1.82	17.7	Retained
LBP mean	97.14	2.24	93.8	Retained
Statistical entropy	95.18	2.78	68.6	Retained
LBP variance	92.14	3.29	100.0	Retained
Angular variance	91.85	3.92	72.4	Retained
GLCM entropy	89.82	4.31	63.5	Retained
LBP entropy	84.81	5.73	100.0	Retained
GLCM homogeneity	81.96	6.27	59.9	Retained
Statistical uniformity	79.52	7.38	172	Excluded
Statistical mean	79.11	13.80	1186	Excluded
Statistical variance	68.51	10.87	611	Excluded
Statistical smoothness	65.24	13.35	779	Excluded
Dominant orientation	62.44	11.24	211	Excluded
GLCM correlation	59.88	20.43	3459	Excluded
GLCM maximum prob.	58.21	20.80	971	Excluded
Radial variance	54.40	11.03	73.2	Excluded
Statistical third moment	52.38	36.57	3534	Excluded
GLCM energy	48.04	41.78	1937	Excluded
Radial mean	47.38	12.87	94.6	Excluded
Fundamental period	42.50	15.44	138	Excluded
LBP uniformity	41.99	36.24	1125	Excluded
GLCM contrast	34.70	24.55	397	Excluded
Dominant frequency	29.05	25.18	241	Excluded
Spectral energy	11.61	27.01	191	Excluded

mean and variance, GLCM homogeneity and the two angular spectral moments. Entropy aggregates over the whole intensity distribution and is consequently insensitive to the addition or removal of a thin boundary shell.

Second, the descriptors that fail are those dominated by

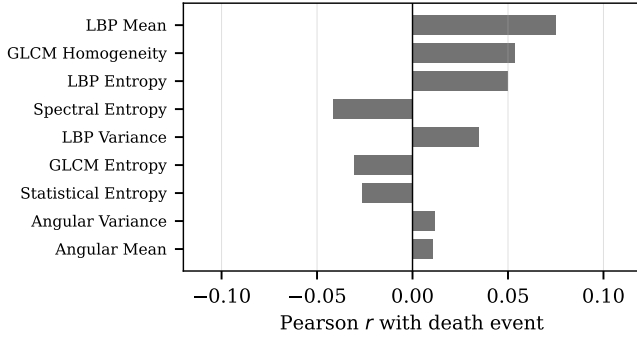


Fig. 5. Pearson correlation between each retained descriptor and the death indicator across the cohort. All nine values lie within ± 0.08 .

extreme values or by a single spectral peak. Spectral energy is stable in only 11.6 per cent of comparisons, and dominant frequency in 29.0 per cent, because a one-voxel change to the mask changes the region support and hence the location of the spectral maximum. GLCM contrast, GLCM energy and the statistical third moment show maximum changes of several hundred to several thousand per cent, which is the behaviour of a quantity driven by a small number of voxel pairs.

Third, and most consequentially for the literature, several descriptors that are routinely reported in radiomic studies—GLCM contrast, GLCM energy, GLCM correlation, first-order mean and variance—do not survive a one-voxel contour perturbation on this cohort.

B. Univariate association with outcome

Before any classifier was trained, the Pearson correlation between each retained descriptor and the death indicator was computed (Fig. 5). The largest absolute correlation is 0.075 for LBP mean, and all nine lie within ± 0.08 . No single retained descriptor carries an appreciable linear association with the outcome. This result anticipates and largely explains the classification performance reported below.

C. Classification performance

Table V reports out-of-fold performance for all six classifiers on the identical set of 419 patients, and Fig. 6 presents the same values graphically. The receiver operating characteristic curves are shown in Fig. 7.

The results must be read against the majority-class rate of 0.598. No classifier exceeds it. The linear support vector machine attains the highest accuracy at 0.597, but does so by predicting class 0 for almost every patient: its specificity is 0.916 and its sensitivity 0.119, and in fold 4 it predicted class 0 for all 84 validation patients, yielding sensitivity exactly zero and specificity exactly one. The random forest gives the most balanced operating point, with sensitivity 0.506 and specificity 0.641, at the cost of accuracy below the majority rate.

The area under the ROC curve is the more informative summary here because it is insensitive to the decision threshold. Its range across the six classifiers is 0.528 to 0.595,

TABLE V
OUT-OF-FOLD CLASSIFICATION PERFORMANCE, 419 PATIENTS,
IDENTICAL FIVE-FOLD PARTITION

Classifier	Acc.	Sens.	Spec.	AUC
Minimum distance	0.568	0.524	0.598	0.585 [†]
Gaussian Bayes	0.570	0.393	0.689	0.595 [†]
Linear SVM	0.597	0.119	0.916	0.528
RBF SVM	0.580	0.173	0.853	0.536
Random forest	0.587	0.506	0.641	0.581
Small CNN	0.578	0.191	0.837	0.566
Majority class	0.598	0.000	1.000	0.500

[†]Computed in this work from the stored out-of-fold decision scores (signed distance difference and log-likelihood difference respectively); the original stage outputs report no AUC for these two classifiers.

TABLE VI
AGGREGATED OUT-OF-FOLD CONFUSION MATRICES

Classifier	TN	FP	FN	TP
Minimum distance	150	101	80	88
Gaussian Bayes	173	78	102	66
Linear SVM	230	21	148	20
RBF SVM	214	37	139	29
Random forest	161	90	83	85
Small CNN	210	41	136	32

that is, between 2.8 and 9.5 percentage points above chance. The two from-scratch classifiers achieve the highest values, 0.585 for minimum distance and 0.595 for Gaussian Bayes, which is notable given that they are the simplest models in the comparison and use all nine descriptors without any per-fold selection.

D. Confusion structure

Table VI gives the aggregated out-of-fold confusion matrices. The pattern is consistent: the classifiers that achieve high accuracy do so by collapsing towards the majority class. The linear support vector machine recovers only 20 of 168 two-year survivors; the random forest recovers 85 but misclassifies 90 of 251 non-survivors in exchange.

E. Fold-level variability

Fig. 8 shows accuracy by validation fold and the sensitivity–specificity operating point of each classifier in each fold. Two features of this figure matter for interpretation. First, fold 4 is systematically easier for every classifier: accuracies there range from 0.595 to 0.702, whereas fold 1 accuracies range from 0.494 to 0.612. The spread across folds is therefore comparable in magnitude to the spread across classifiers, which means that a single held-out split would have supported almost any ranking of the six models.

Second, the operating points cluster along the anti-diagonal. Classifiers trade sensitivity for specificity along a line rather than moving outward towards the upper-right corner, which is the visual signature of an absent discriminative signal rather

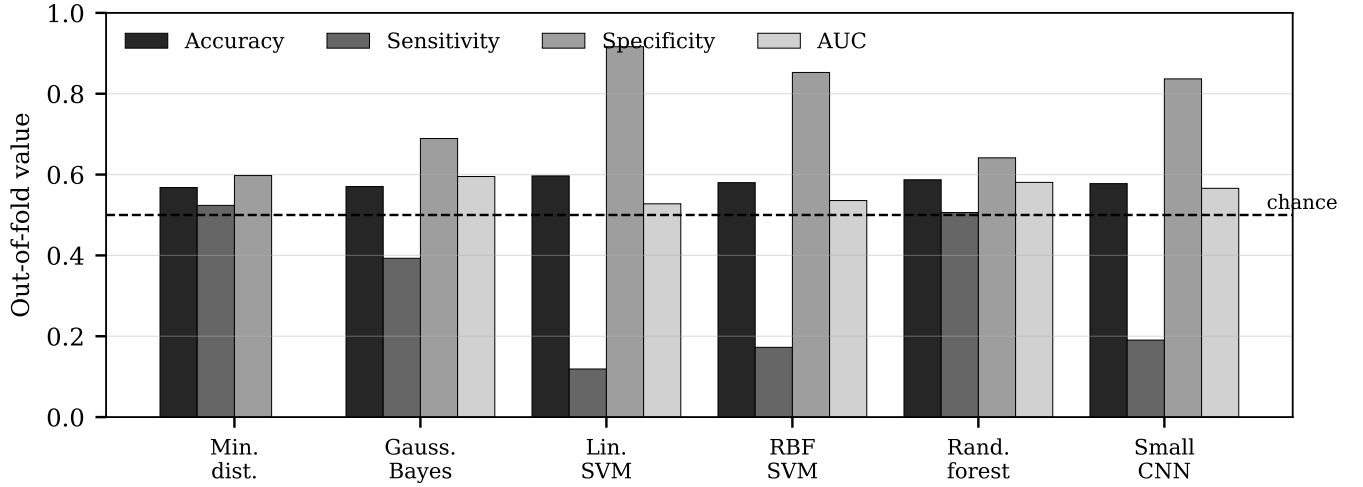


Fig. 6. Out-of-fold accuracy, sensitivity, specificity and area under the ROC curve for the six classifiers. The dashed line marks chance level for AUC and, coincidentally, is close to the majority-class accuracy of 0.598.

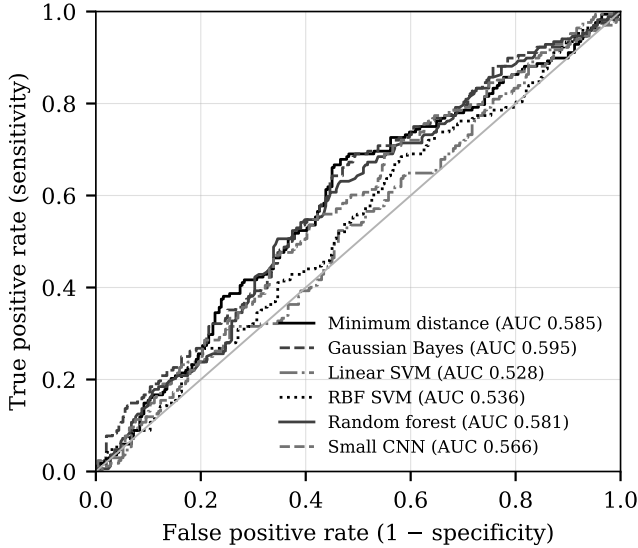


Fig. 7. Out-of-fold receiver operating characteristic curves. All six curves lie close to the diagonal; the best area under the curve is 0.595.

TABLE VII
ACCURACY BY VALIDATION FOLD

Classifier	F1	F2	F3	F4	F5
Minimum distance	0.494	0.506	0.571	0.690	0.578
Gaussian Bayes	0.494	0.518	0.571	0.702	0.566
Linear SVM	0.612	0.566	0.548	0.595	0.663
RBF SVM	0.553	0.542	0.548	0.643	0.614
Random forest	0.506	0.578	0.595	0.667	0.590
Small CNN	0.600	0.542	0.512	0.619	0.614

than of a poorly chosen threshold. Table VII gives the per-fold accuracies numerically.

V. DISCUSSION

A. Interpretation

The central result is negative: nine stability-filtered texture descriptors extracted from the GTV-1 region do not predict two-year survival in the Lung1 cohort at a level of practical use. The best area under the ROC curve is 0.595 and no classifier beats a majority-class predictor on accuracy.

This outcome is coherent across three independent lines of evidence in the data, which is what makes it credible rather than merely disappointing. The univariate correlations are all below 0.08 in absolute value; the ROC curves lie close to the diagonal for models spanning a wide range of capacity, from a nearest-mean rule to a 500-tree ensemble; and the operating points move along the sensitivity–specificity anti-diagonal rather than away from it. A pipeline defect would be unlikely to produce all three patterns simultaneously and consistently.

It is also informative that increasing model capacity does not help. The random forest and the convolutional network do not outperform the minimum-distance classifier in AUC terms. When capacity is irrelevant, the limitation is normally in the representation rather than in the decision rule.

B. Why the descriptors may be uninformative here

Several explanations are consistent with the observations. The stability filter removed sixteen of twenty-five descriptors, and it is possible that some discarded descriptors were both unstable and genuinely informative; stability and prognostic value are distinct properties, and the filter optimises only the first. The retained set is also strongly redundant—six of the nine are entropy-like or LBP-derived—so the effective dimensionality of the descriptor space is smaller than nine.

The acquisition parameters of Lung1 are heterogeneous across patients in reconstruction kernel, slice thickness and manufacturer. Texture descriptors are known to be sensitive

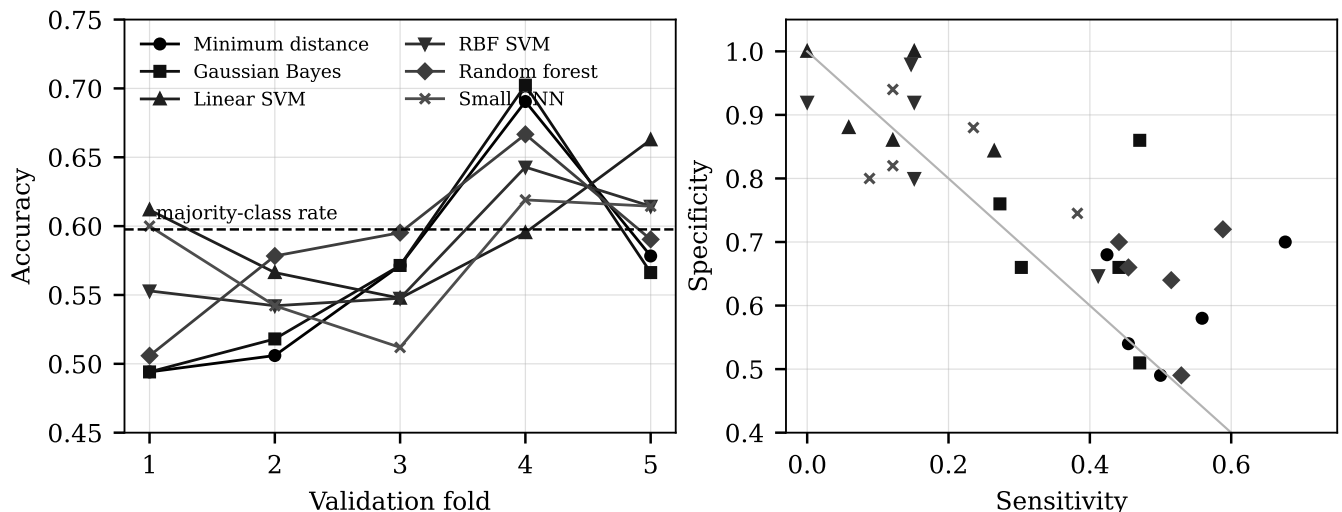


Fig. 8. Left: out-of-fold accuracy by validation fold for each classifier; the dashed line is the majority-class rate. Right: the sensitivity–specificity operating point of each classifier in each fold. Points cluster along the anti-diagonal rather than moving towards the upper-right corner.

to these factors [9], and no resampling to isotropic voxels or intensity discretisation harmonisation was applied, so a portion of the descriptor variance is acquisition variance rather than biological variance.

Finally, the endpoint itself is demanding. Two-year survival in non-small-cell lung cancer is influenced by stage, performance status, treatment received and comorbidity, none of which was included in the descriptor vector used here. A radiomic-only model is being asked to recover an outcome that is only partly determined by tumour appearance.

C. Limitations

The limitations that bound these conclusions are as follows.

No clinical baseline. The clinical variables were merged and quality checked, but no clinical-only model (stage, age, histology) was fitted. Without it, the marginal value of the radiomic descriptors above routinely available clinical information cannot be quantified. This is the single most useful addition that future work could make.

No confidence intervals. Point estimates are reported without bootstrap confidence intervals. Given 419 patients and five folds, the uncertainty on an AUC of 0.595 is wide enough that it may not be distinguishable from chance, and the fold-level spread in Fig. 8 suggests it would be.

Fixed stability thresholds. The ten-per-cent and eighty-per-cent thresholds were fixed a priori and never varied. A sensitivity analysis over these thresholds would show how much the retained set, and hence the result, depends on them.

Shape descriptors not carried forward. The boundary, regional and three-dimensional shape descriptors were implemented and validated but were not included in the classification descriptor vector. Tumour volume in particular is an established prognostic factor in this disease, and its absence from the model is a deliberate scope decision, not evidence that it lacks value.

Convolutional network scope. As stated in Section III, the network operates on a 3×3 arrangement of the nine retained descriptors, not on image patches. It therefore tests the descriptor representation, not the hypothesis that a deep model reading raw tumour images could do better.

Single cohort, single partition. All results derive from one institution-level public cohort and one fixed cross-validation seed. No external validation was performed.

VI. CONCLUSION

A complete radiomic pipeline was implemented from first principles and applied to the 422-patient Lung1 cohort, covering DICOM ingestion, RTSTRUCT-based tumour mask reconstruction, boundary and shape descriptors, four families of texture descriptors, mask-perturbation stability filtering, and six classifiers evaluated on a single frozen patient-level cross-validation partition.

Two results are worth carrying forward. The first is methodological: only nine of twenty-five texture descriptors survived a one- and two-voxel contour perturbation at a ten-per-cent tolerance, and among those eliminated were several descriptors reported routinely in the radiomics literature. Stability testing is not an optional refinement; it changes which descriptors are admissible.

The second is empirical and negative. On the retained descriptor set, no classifier exceeded the majority-class accuracy of 0.598, and the best area under the ROC curve was 0.595. Reported honestly, this quantifies how much prognostic signal these descriptors carry for this endpoint in this cohort: very little. The natural next steps are a clinical-only baseline for comparison, bootstrap confidence intervals on every reported metric, and the reintroduction of volume and shape descriptors into the classification vector.

REPRODUCIBILITY

Every number in this paper is produced by a numbered stage script that writes its own result files. The implementation is available at <https://github.com/Ceceliaal-qaisi/NSCLC-Radiomics-GitHub>. The imaging and clinical data are distributed by The Cancer Imaging Archive under the NSCLC-Radiomics collection and are not redistributed here.

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